

< Programming Assignment #1 >

Submission: LMS (learning.hanyang.ac.kr)

- **Until April 7 (Tuesday), 23:59 PM.**

1. Environment

- OS: Windows, Mac OS, or Linux
- Language: Python (any version is fine, but python3 is recommended)

2. Goal: to find **association rules** using the **Apriori** algorithm

- step 1: find frequent itemsets using Apriori
- step 2: for each frequent itemset, find association rules

3. Requirements

The program must meet the following requirements:

- File name: studentID.py (**not ipynb, Zip, etc**)
 - Only put your student ID. Do NOT put any other information.
 - Example 1: 2008011666.py
- I will execute your code with **three** arguments: **minimum support (%)**, **input file name**, **output file name**
 - Example: apriori.py 5 input.txt output.txt
 - This means that **minimum support = 5%**, input file name = **'input.txt'**, output file name = **'output.txt'**
 - I will test your assignment using several different minimum support percentage values.

- Input file format (.txt)

`[item_id]\t[item_id]\n`

`[item_id]\t[item_id]\t[item_id]\t[item_id]\t[item_id]\n`

`[item_id]\t[item_id]\t[item_id]\t[item_id]\n`

- Each line: transaction
- `item_id` is a numerical value
- There is no duplication of items in each transaction
- Example:

18	2	4	5	1	
1	11	15	2	7	16
2	1	16			
15	7	6	11	18	9
11	2	13	4		

- Output file format (.txt)

`{item_set}\t{associative_item_set}\tsupport(%) \tconfidence(%) \n`

`{item_set}\t{associative_item_set}\tsupport(%) \tconfidence(%) \n`

- `{item_set}\t{associative_item_set}`: association rules with minimum support (Minimum confidence is **not** considered. Only consider minimum support)

- $\{item_set\} \rightarrow \{associative_item_set\}$
- Use **braces** ({ }) to represent item sets: $\{item_id, item_id, \dots\}$ (Important!!)
- **Support**: probability that a transaction contains $\{item_set\} \cup \{associative_item_set\}$
- **Confidence**: conditional probability that a transaction having $\{item_set\}$ also contains $\{associative_item_set\}$
- The order of lines in the output file is not important.
- The value of support and confidence should be rounded to **two decimal places**.
 - e.g., 24.631 should become 24.63.
- Result example (it is obtained from a minimum support of 5%:

```

      .
      .
      .
{12,16} {13}      5.20      38.81
{13,16} {12}      5.20      37.68
{1}      {3,8,16}      9.40      31.54
{3}      {1,8,16}      9.40      31.33
{1,3}    {8,16}      9.40      87.04
{8}      {1,3,16}      9.40      20.80
{1,8}    {3,16}      9.40      61.04
{3,8}    {1,16}      9.40      36.43
{1,3,8}  {16}        9.40      97.92
{16}     {1,3,8}      9.40      22.17
{1,16}   {3,8}        9.40      58.02
{3,16}   {1,8}        9.40      37.30
{1,3,16} {8}          9.40      97.92
{8,16}   {1,3}        9.40      31.13
{1,8,16} {3}          9.40      81.03
{3,8,16} {1}          9.40      39.17

```

- This is how to generate association rules based on a frequent itemset:
 - Assume that we have a **frequent itemset** {1, 2, 3}.
 - From this itemset, we can generate the following association rules:
 - {1}->{2,3}
 - {1,2}->{3}
 - {1,3}->{2}
 - {2}->{1,3}
 - {2,3}->{1}
 - {3}->{1,2}
 - As you can see, in any $A \rightarrow B$, $A \cup B$ must be {1, 2, 3}, and there must be no intersection among A and B.
 - The order of the association rules, and the order of items in each itemset is not important.
 - And whether there is a space between an item and comma(,) is also not important.
- Note: Please make sure to match the output format! If the format is not correct, you can't get any score.

4. Submission

- Please make a **single, runnable .py file** and upload it on LMS.

5. Penalty

- **0 score**: Late submission, Library violation
 - Some libraries include Apriori algorithm implementations, but these libraries are **strictly prohibited**! Only basic libraries like numpy, pandas, should be used, and **the libraries which are directly related to and have implementations of the assignments must NOT be used**.
- **F grade**: Plagiarism